School of Information Technology, Engineering, Mathematics and Physics

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CS241 – Software Design & Implementation
Semester 2, 2024
Mode: Face to Face

Short Test 2
Date: Tuesday 8 October 2024, Time: 11.00 – 11.50am (FJT)

Time Allowed: 50mins
30 marks (Weight: 10%)

Student ID: _____
Student Name: _____

LEARNING OUTCOMES:

- Develop software design document
- Develop a simple software system based on the design document

INSTRUCTIONS

1) This exam has three sections:
   a) Section A: Short-answer questions (25 marks)
   b) Section B: Programming (5 marks)
2) Answer all questions in this booklet. All questions are compulsory.
3) The total number of pages including this cover sheet is 6.

~ALL THE BEST~

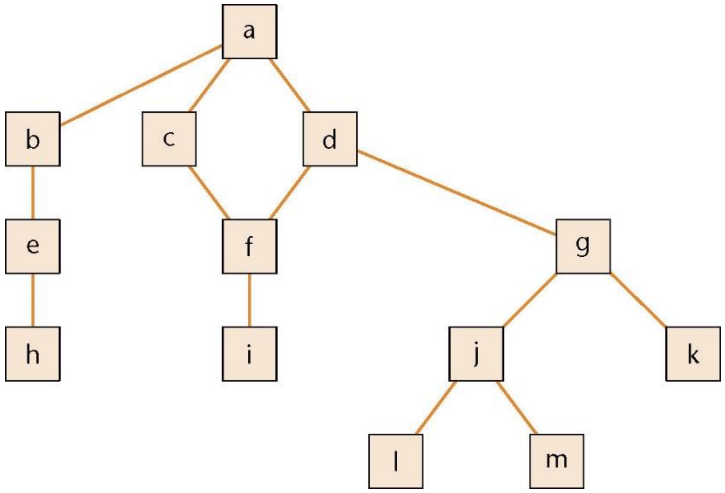
Section A: Short Answer Questions *(25 Marks)*

1. Why are both Software inspections and Software testing important? (2 marks)
   static testing is crucial for early defect detection in software documentation and code, while testing verifies the software's behavior and performance during execution. Together, they provide a comprehensive approach to ensuring software quality and reliability. Both types of testing are essential components of a robust software testing strategy.

2. What is Unit testing? (1 mark)
   - Unit testing is the process of testing individual components in isolation.

3. What is Software Architecture? (1 mark)
   - **Software Architecture** refers to the high-level structure of a software system, representing the fundamental organization of its components and their relationships.
   - It defines how the different parts of the system, such as modules, services, and components, interact with each other to fulfill the system's objectives.
   - The architecture outlines the system's structure, behavior, and the interactions between its components, providing a blueprint for the design and development process.

4. Consider the following system with 13 modules and use it to answer questions a - e.



a) What is a logical artefact? (1mark)

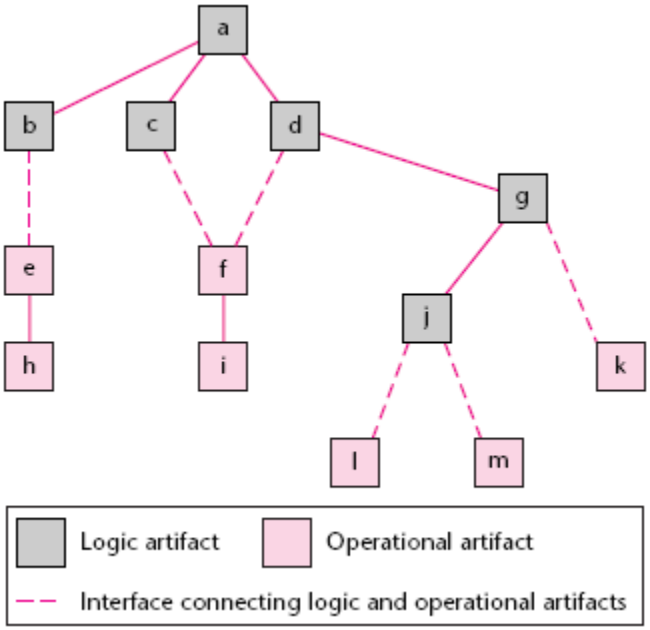
   Logical artefact depicts the decision-making flow of control aspects of a product.

b) What is an operational artefact? Give an example. (1mark)
   Operational artefact performs the actual operations of the product.

c) Observe the above interconnection diagram depicting the organisation of various modules in a software design. Identify the logical and operational artefacts from the diagram. (2 marks)

Logical artefacts: a, b, c, d, g j
Operational artefacts: e, f, h, i, k, l, m

d) Redraw the interconnection diagram to illustrate the sandwich integration in software implementation. (3 marks)



e) Describe the sandwich integration strategy and give one advantage of this method. (3 mark)

Logic artifacts are integrated top-down. Operational artifacts are integrated bottom-up. Finally, the interfaces between the two groups are tested. // 2 marks

Adv // 1 mark

- Major design faults are caught early
- Operational artifacts are thoroughly tested
- They may be reused with confidence
- There is fault isolation at all times

5. You are assigned in your current sprint to create a method that returns the grade of students based on the total marks they have received following the given rule:

A	B	C	D	E
80-100	65-79	50-64	49-30	29-0

*Assume the argument "Mark" is of type int and the method returns "X" for invalid marks

After implementing the method you are required to create unit test cases before your pull request can be processed.

Using Equivalence Partitioning, state all valid and invalid partitions. (3 marks)

Valid: 0<=mark<=29, 30<=mark<=49, 50<=mark<=64, 65<=mark<=79, and 80<=mark<=100 // 2 marks Invalid: Mark<0 and mark >100 // 1 marks

6. You are developing a web application that allows users to input their ages. The application restricts the allowed age range to 18 to 65 years. State the test inputs that would be selected using three-value BVA to test this age input functionality (3 marks).

-17,18, 19 | 64, 65, 66 since it restricts

7. Find the errors in the following code and write in the provided space: (2 marks)

if (t > 2 && m < 5) {
  if (t < 1)
  x = m * t;
}
	for (j=5;j > 5;j--)
	  total = total + value[j];

If (t<1) will never be executed – 1mark	For loop never executed -1mark

8. What is the purpose of the Model in Model-View-Controller (MVC) design pattern and give one advantage of using MVC pattern? (3 marks)

The Model represents the application's data and business logic. It encapsulates the data and provides methods to manipulate and interact with it. The Model is responsible for maintaining the integrity and consistency of the data. // 1marks

Adv // 1 mark

- Separation of Concerns: It promotes a clear separation of responsibilities between data handling (Model), user interface (View), and user input handling (Controller).
- Maintainability: Changes to one component (e.g., the View) can often be made without affecting the others, making the codebase easier to maintain.
- Testability: Each component can be tested independently, which simplifies unit testing.

- Scalability: MVC facilitates scalability by allowing the addition of new Views or Controllers without affecting the core business logic (Model).

Section B: Programming *(5 Marks)*

The multi-threaded program contains two or more parts that can run concurrently, and each part can handle a different task at the same time making optimal use of the available resources specially when your computer has multiple CPUs. One way to create multi-threaded program is by Creating a Thread by Implementing a Runnable Interface as shown below:

```
class MessagePrinter implements Runnable {
    private String msg;
    private int repeat;

    public MessagePrinter(String msg, int repeat) {
        this. msg = msg;
        this.repeat = repeat;
    }

    public void run() {
        try {
            for (int i = 1; i <= repeat; i++) {
                System.out.println(Thread.currentThread().getName() + ": " + msg);
                Thread.sleep(300);  // Pausing for 300 milliseconds between prints
            }
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }
}
```

Implementing three threads of `MessagePrinter` in the driver given below. Only write within the main. (5 marks)

```
public class MultiThread {
    public static void main(String[] args) {
       // pass " Good morning" in first threaded, repeat 3 times
      // pass " Good day" in first threaded, repeat 3 times
        Runnable r1 = new MessagePrinter("Good morning", 5);
        Runnable r2 = new MessagePrinter("Good night", 5);

        Thread t1 = new Thread(r1, "MorningThread");
        Thread t2 = new Thread(r2, " DayThread ");

        t1.start();
        t2.start();

    }
```

```
        // 1 mark each
```

THE END

EXTRA ANSWER SHEET – USE ONLY WHEN NEEDED

- Mark the question number properly if used.